



Algebra: Simplifying Radical Expressions

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DIRECTIONS

Simplify each radical expression. Factor the radicand into a product containing the largest perfect-square (or perfect-cube) factor, then take that factor out of the radical. Assume all variables represent nonnegative real numbers.

1. Simplify:

$$\sqrt{32}$$

Answer: _____

2. Simplify:

$$\sqrt{48}$$

Answer: _____

3. Simplify:

$$\sqrt{72}$$

Answer: _____

4. Simplify:

$$\sqrt{80}$$

Answer: _____

5. Simplify:

$$\sqrt{x^5}$$

Answer: _____

6. Simplify:

$$\sqrt{9x^2}$$

Answer: _____

7. Simplify:

$$\sqrt{8x^3}$$

Answer: _____

8. Simplify:

$$\sqrt{50x^2y}$$

Answer: _____

9. Simplify:

$$\sqrt{45a^2b^3}$$

Answer: _____

10. Simplify:

$$\sqrt[3]{16}$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES

Items 1–4: numeric square roots — factor out the largest perfect square. Items 5–6: variables only. Items 7–9: mixed numerical and variable radicands. Item 10: cube root — factor out the largest perfect cube.

1. Simplify: $\sqrt{32}$

Answer: $4\sqrt{2}$

$$32 = 16 \cdot 2 \rightarrow \sqrt{32} = 4\sqrt{2}.$$

2. Simplify: $\sqrt{48}$

Answer: $4\sqrt{3}$

$$48 = 16 \cdot 3 \rightarrow \sqrt{48} = 4\sqrt{3}.$$

3. Simplify: $\sqrt{72}$

Answer: $6\sqrt{2}$

$$72 = 36 \cdot 2 \rightarrow \sqrt{72} = 6\sqrt{2}.$$

4. Simplify: $\sqrt{80}$

Answer: $4\sqrt{5}$

$$80 = 16 \cdot 5 \rightarrow \sqrt{80} = 4\sqrt{5}.$$

5. Simplify: $\sqrt{x^5}$

Answer: $x^2\sqrt{x}$

$$x^5 = x^4 \cdot x \rightarrow \sqrt{x^5} = x^2\sqrt{x}.$$

6. Simplify: $\sqrt{9x^2}$

Answer: $3x$

$$\text{Both } 9 \text{ and } x^2 \text{ are perfect squares: } \sqrt{9x^2} = 3x.$$

7. Simplify: $\sqrt{8x^3}$

Answer: $2x\sqrt{2x}$

$$8x^3 = 4x^2 \cdot 2x \rightarrow \sqrt{8x^3} = 2x\sqrt{2x}.$$

8. Simplify: $\sqrt{50x^2y}$

Answer: $5x\sqrt{2y}$

$$50x^2y = 25x^2 \cdot 2y \rightarrow \sqrt{50x^2y} = 5x\sqrt{2y}.$$

9. Simplify: $\sqrt{45a^2b^3}$

Answer: $3ab\sqrt{5b}$

$$45a^2b^3 = 9a^2b^2 \cdot 5b \rightarrow \sqrt{45a^2b^3} = 3ab\sqrt{5b}.$$

10. Simplify: $\sqrt[3]{16}$

Answer: $2\sqrt[3]{2}$

$$16 = 8 \cdot 2 \rightarrow \sqrt[3]{16} = 2\sqrt[3]{2}.$$

